

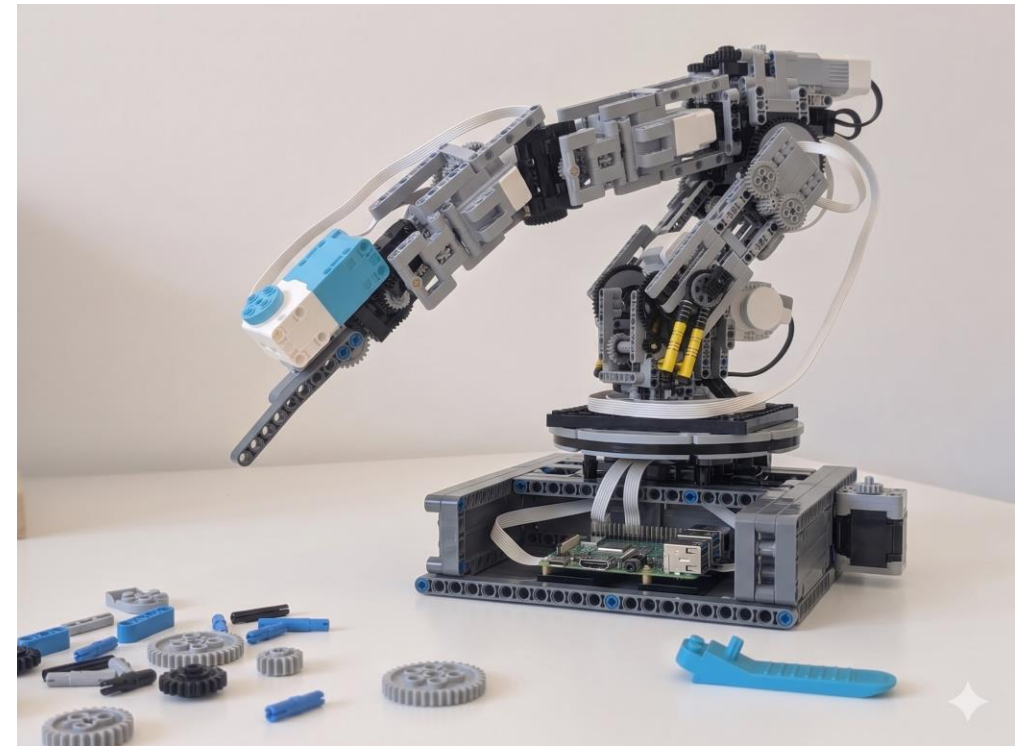
First Milestone (Team 1)

Abdelrahman Elkholy

Andreas Flohr

◆ What is the Project about?

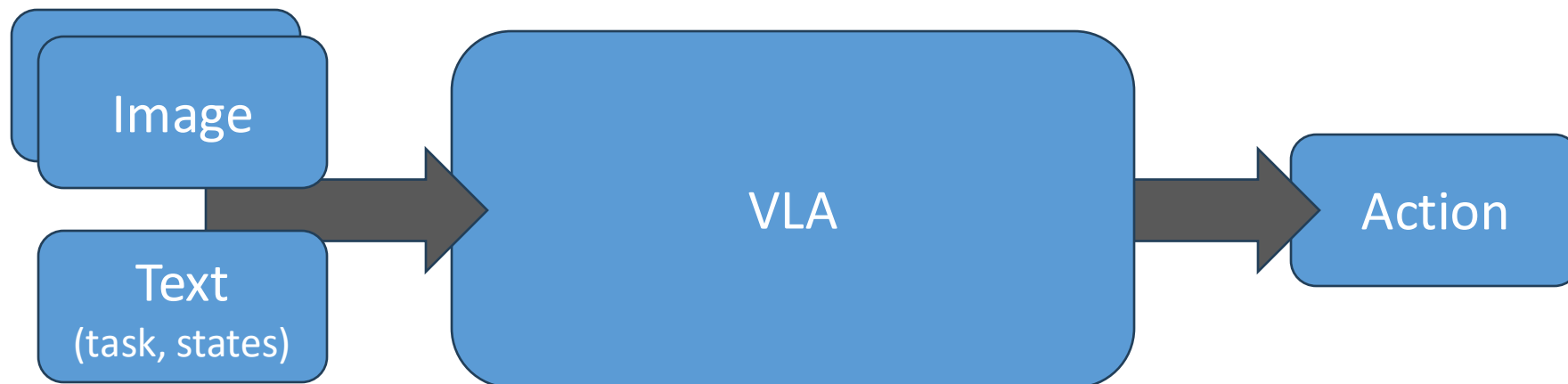
- Build a custom LEGO robot arm
- Perform a specific manipulation task
- No handcrafted algorithms
- Use a Vision-Language-Action model



Prototype: Robot (Nano banana)

◆ Vision Language Action model

- ***A generative end-to-end model that maps visual and natural language inputs directly to robotic control actions.***
- Inputs (Image, Text) → Tokens → Outputs (Robot action)
- Transformer based models: Octo, RT-2, ...



Handcrafted Algorithms vs VLA Models

■ Handcrafted Algorithms

- Build to work in one environment
- Build to do one task
- Execution fails on state mismatch

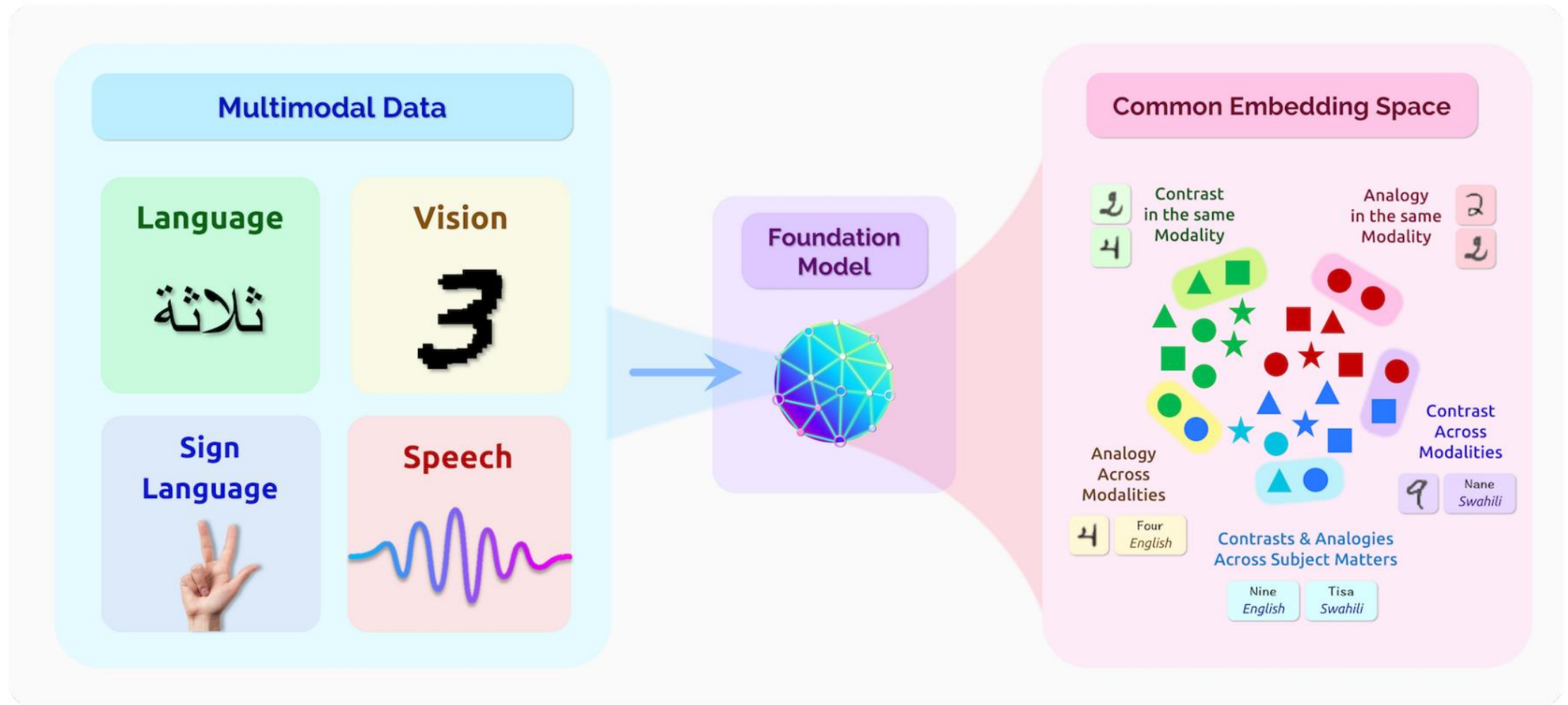
■ VLA models

- Environment independent
- Can adapt to new tasks
- Next state based on last state

◆ Foundation models

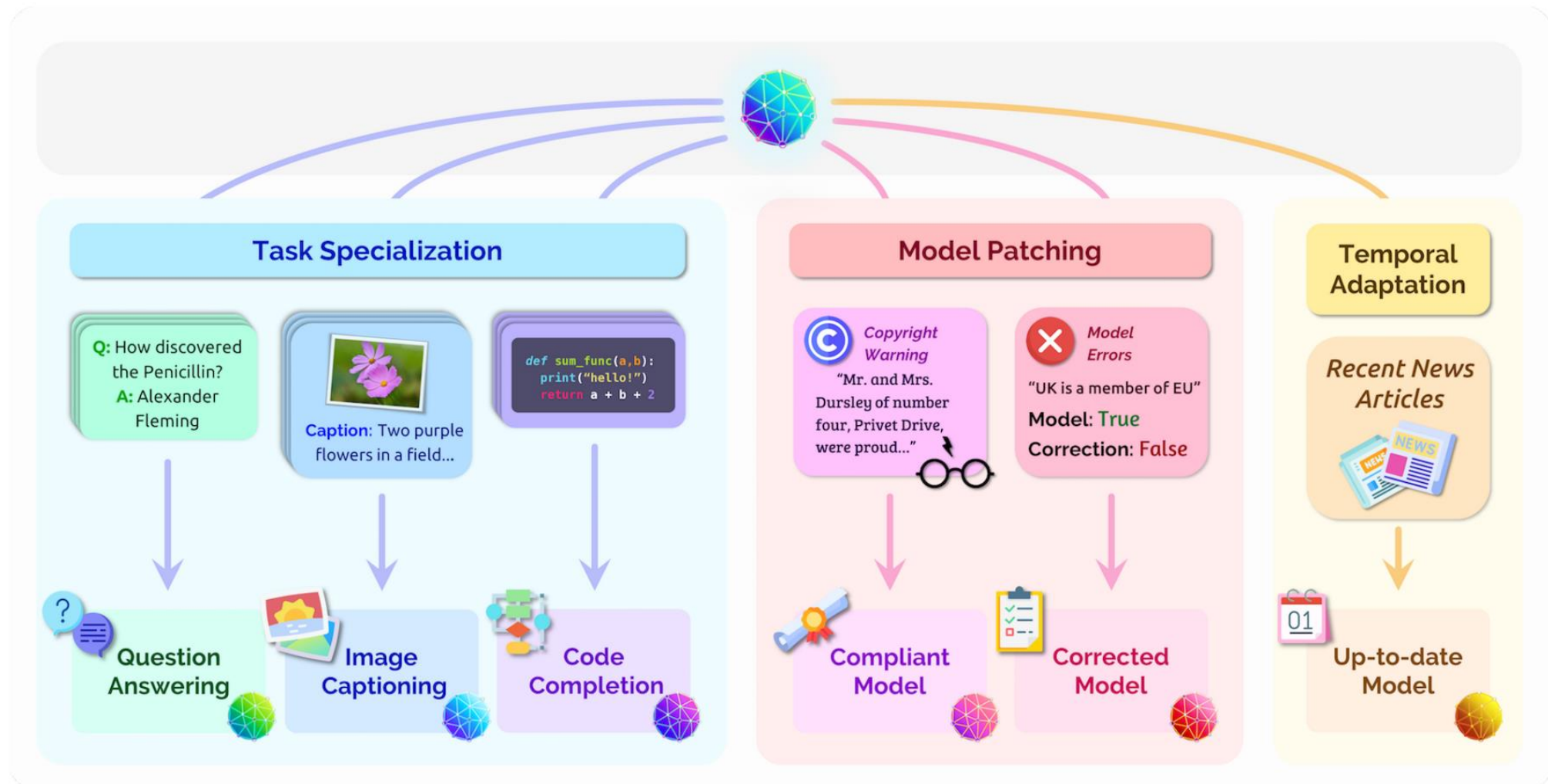
- *A model trained on a vast amount of data that can be adapted (fine-tuned) to a wide range of tasks.*
- **Specialization:** Can be focused on specific goals/topics like robotic.
- **Reuse:** A single base model can be used for many different tasks/goals.
- **Efficiency:** Needs small amount of new data to adapt.
- **Pre-Training:** Learns broad patterns from huge datasets.
- **Fine-tuning:** Uses existing knowledge to learn new tasks quickly.

◆ Pre-training



[4] OtOaRaFM p. 69: various modalities into universal feature space

Fine-tuning



[4] OtOaRaFM p. 85: foundation model is converted into an adapted model

◆ Why not train from scratch?

■ Training from scratch

- Requires millions of demonstrations and months of GPU training
- Overfits on small datasets, poor generalization

■ Finetuning a pretrained model (Octo)

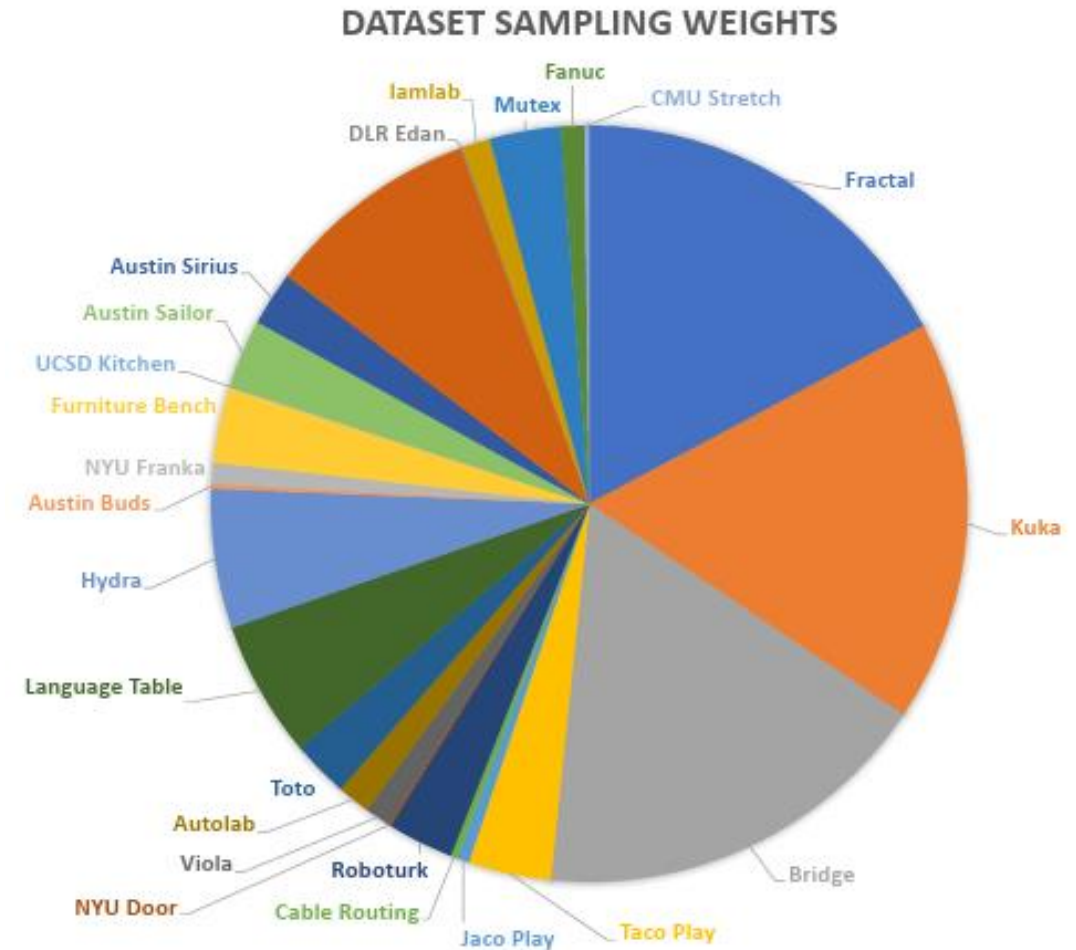
- ~ 100 demonstrations, about 5 hours on a single GPU
- Transfers knowledge from 800k trajectories across 22 robots

Method	Avg Success Rate
From Scratch	20%
Octo (Finetuned)	72%

[1] Octo Model Team et al., 2024, Table 1

◆ Octo Overview

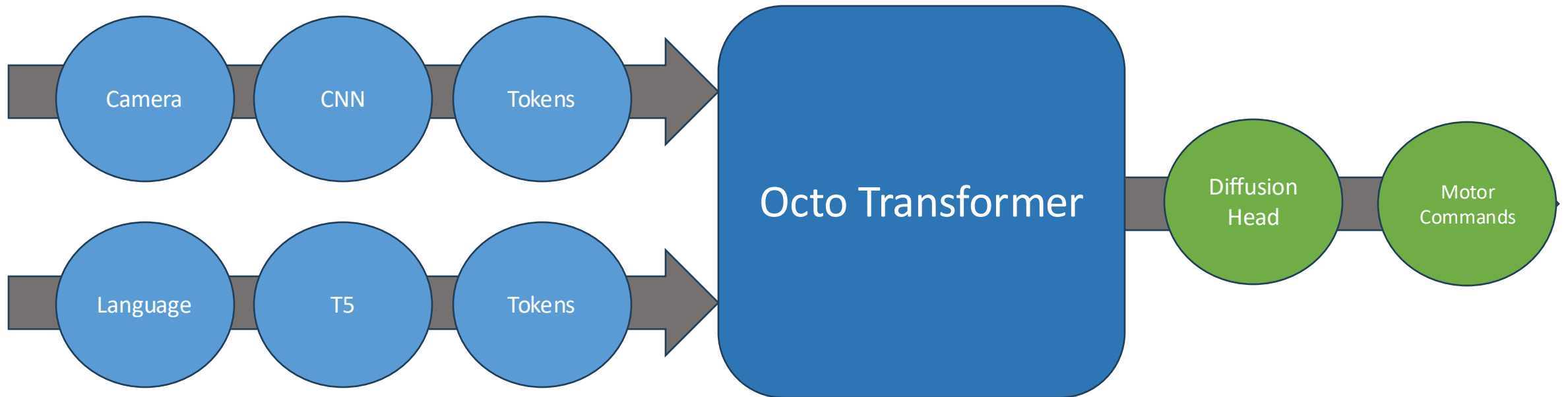
- Open-source generalist robot policy (UC Berkeley, 2024)
- Two sizes: Octo-Small (27M) and Octo-Base (93M params)
- Outperforms RT-1-X, performs comparably to RT-2-X (55B params)



[1] Octo Model Team et al., 2024, Fig. 3

◆ Octo Architecture

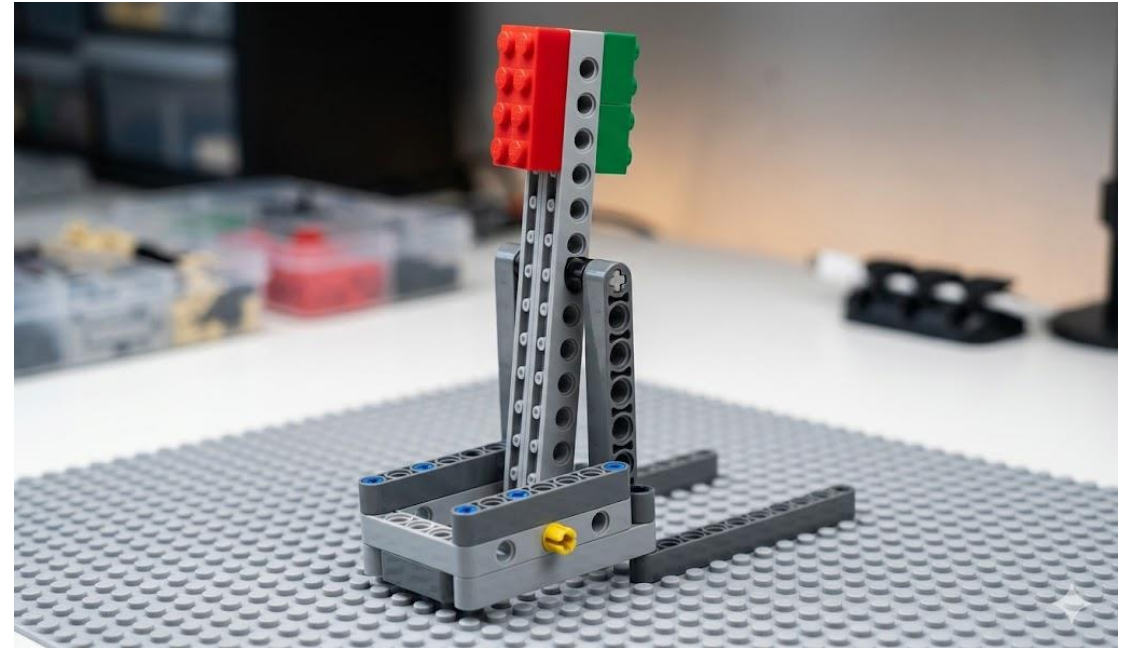
- Camera frames and language instructions are jointly processed by Octo to produce motor commands



CNN = Convolutional Neural Network · T5 = Pretrained Language Model · Diffusion Head = Action Decoder

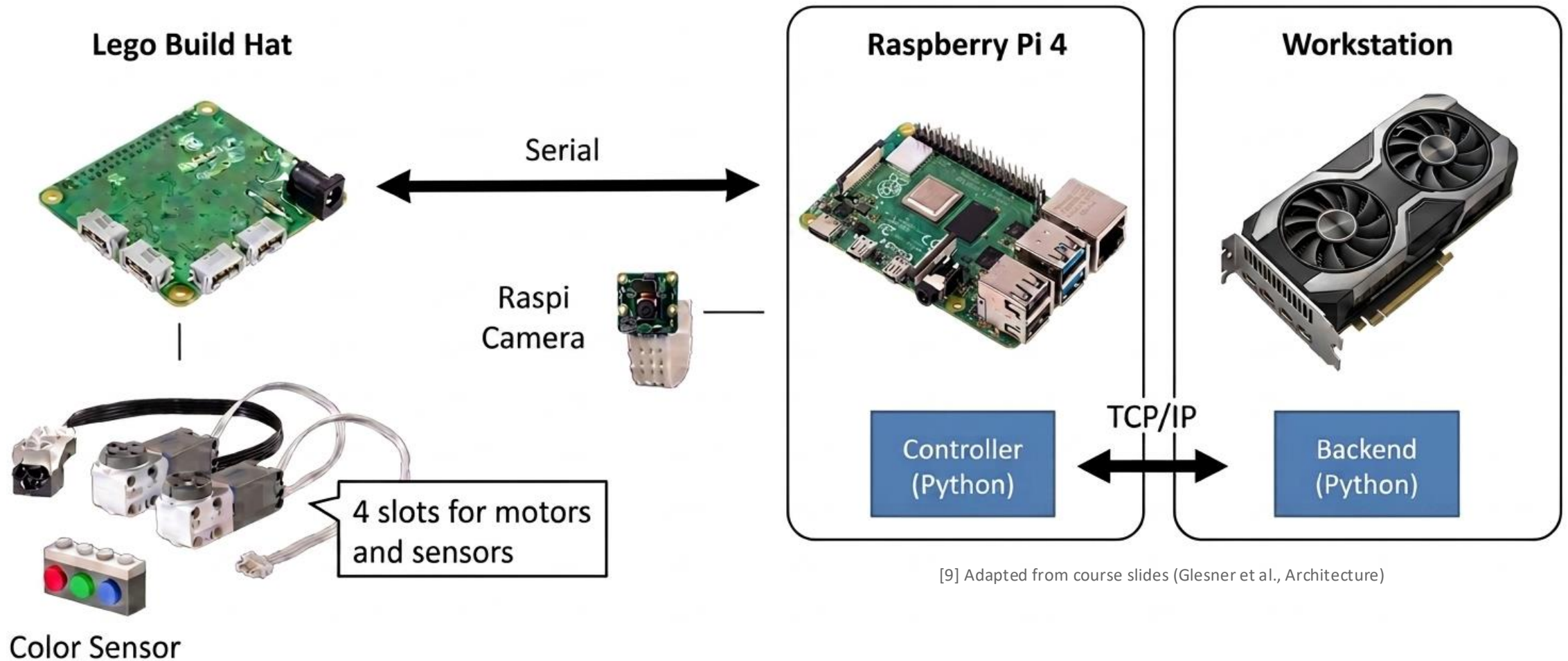
◆ Project Goal

- Train a LEGO arm to flip a marked lever using a finetuned Octo model
- Color sensor reads lever state to determine which direction to flip

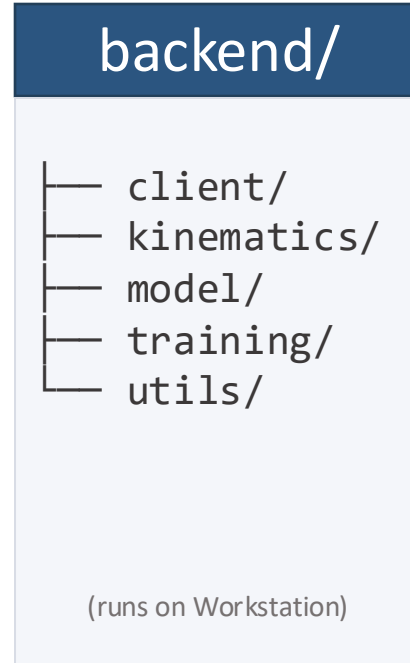
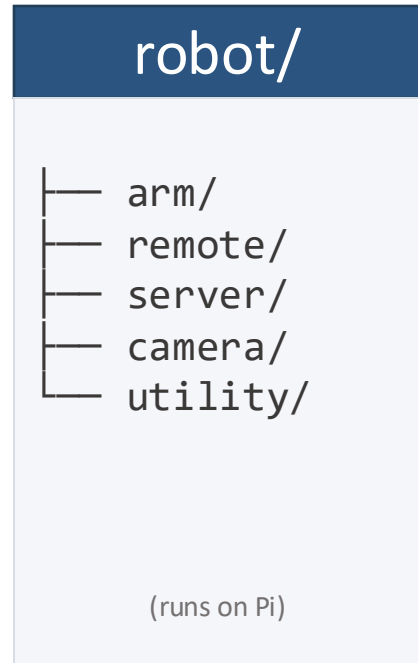


Prototype: lever with red/green sides

◆ System Architecture



[9] Adapted from course slides (Glesner et al., Architecture)



Key libraries & algorithms

buildhat

official Python library: handles serial protocol + PID motor control

octo(JAX)

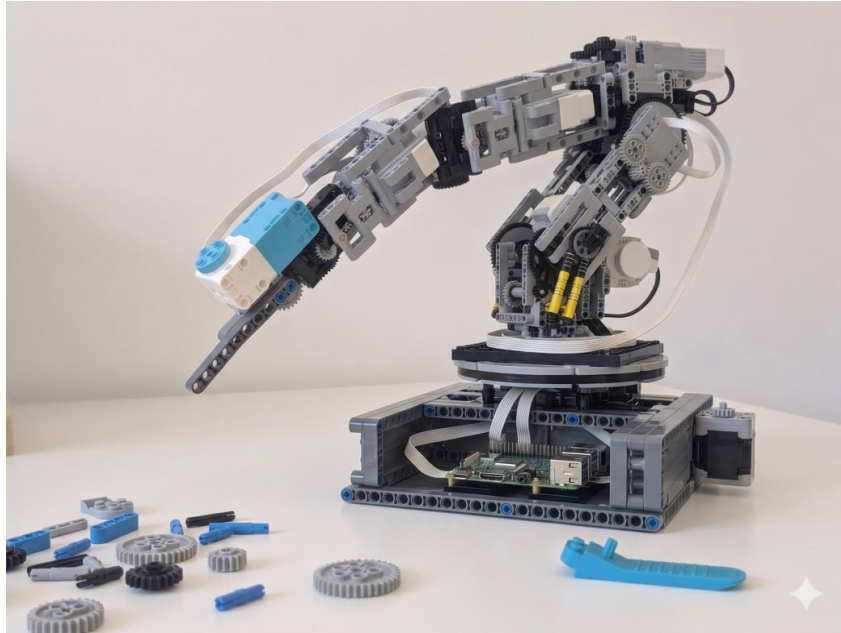
pretrained Vision-Language-Action model for inference

pinocchio

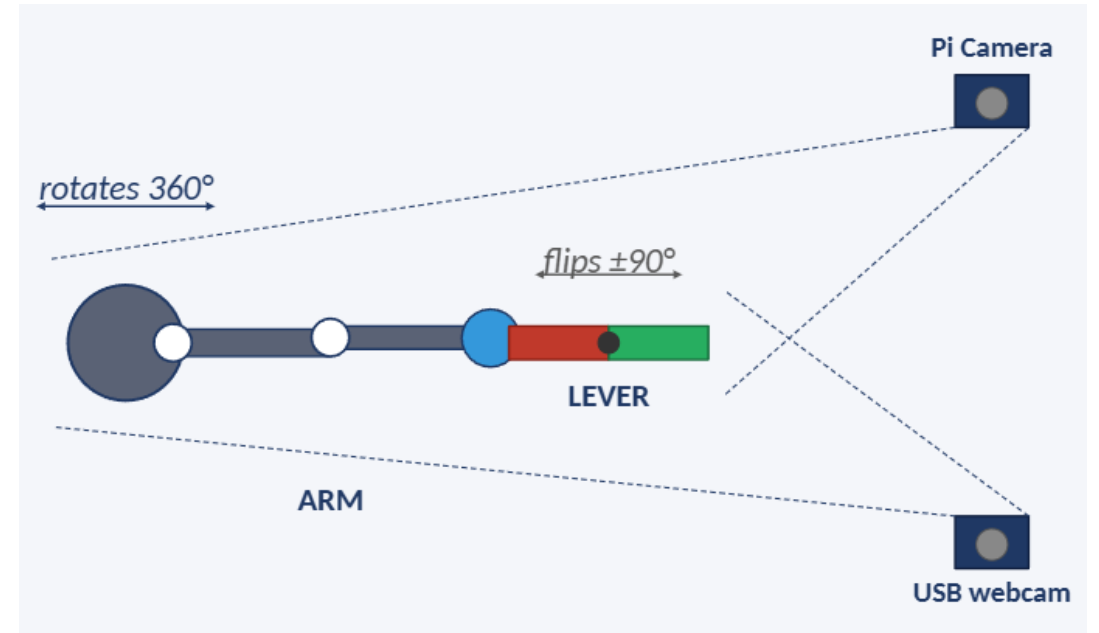
inverse kinematics: converts end-effector pose to joint angles

Unified Python PEP 8 stack across the system

◆ Robot Design & Environment



Prototype: Robot (Nano banana)



Workspace layout, top down view

- **Robot Design:**
Rotating base + shoulder and waist joint with a simple end-effector for pushing the lever
- **Environment:**
Fixed workspace with cameras

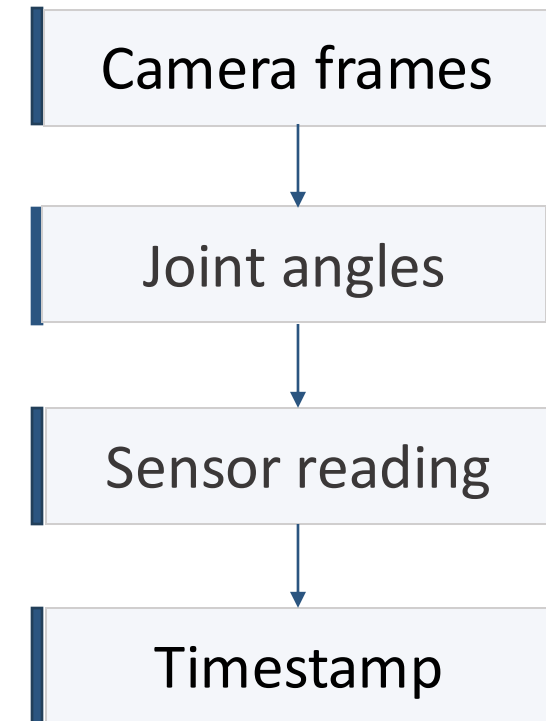
◆ Data Collection - Approach

■ Method

- Teleoperation via Keyboard or Joystick
- Rotate operators to add motion variety

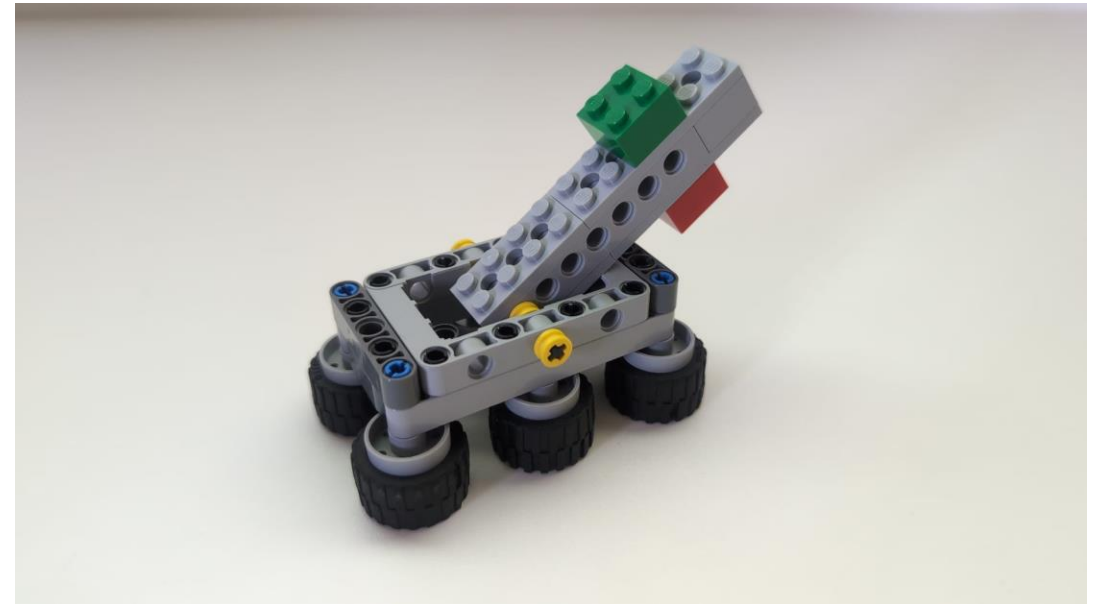
■ Recording Setup

- 5Hz capture rate, 640x480 resolution
- No reset between Trajectories



◆ Data Collection - Quality & Augmentation

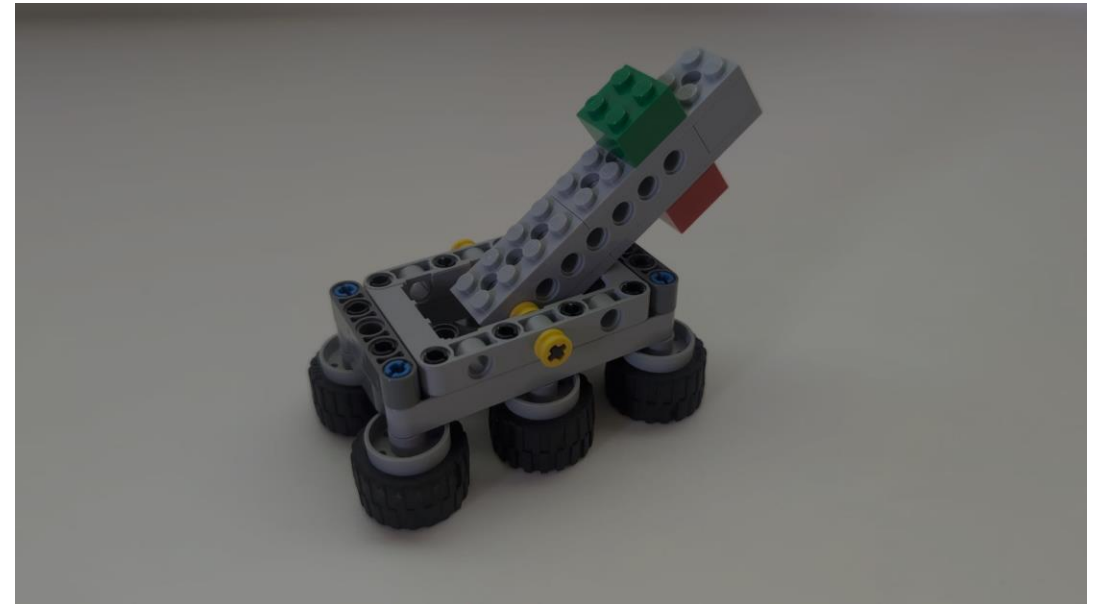
- Task conditioning via color sensor for multitask generalization
- Brightness variation, camera randomization every couple of demos
- Synthetic data as fallback



Original frame captured during teleoperation

◆ Data Collection - Quality & Augmentation

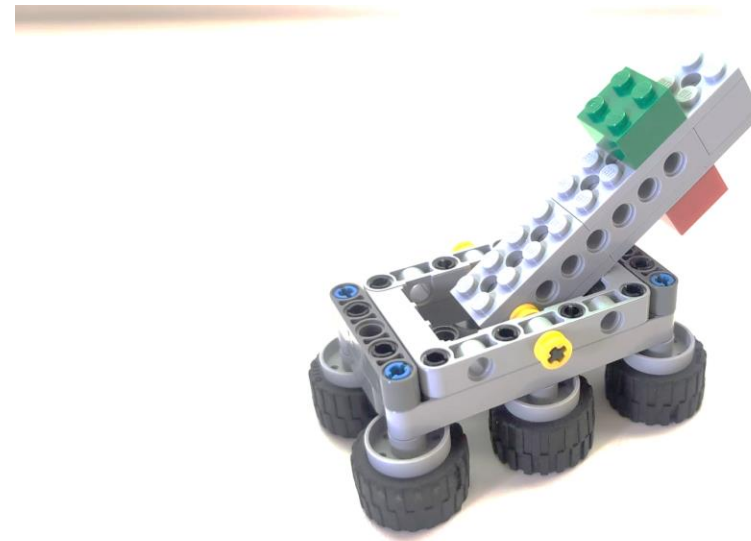
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Darkened to handle bright lighting conditions

◆ Data Collection - Quality & Augmentation

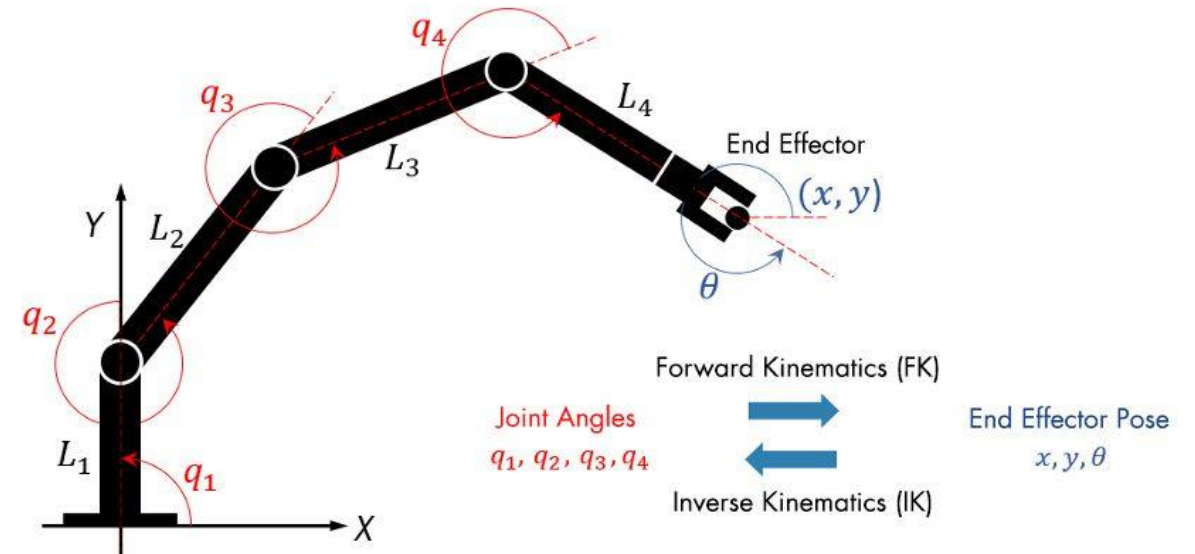
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Brightened to handle bright lighting conditions

Kinematics

- Kinematics types:
 - Forward Kinematics
 - Inverse Kinematics (what we need)
- Python library: pinocchio
 - Open source
 - For dynamic computation (C++)
 - Focusing on robotic



[8] Mathworks: Kinematics



Project Requirements

Functional	Non-Functional
Object Detection	Reliability
Object Localization	Maintainability
Detect Errors	Scalability
Agile Motion	Safety (no harm to environment & user)
Adaptive Lever Flipping	
Receive & Follow User Commands	

◆ Safety Requirements

- **Hardware (Robot):**

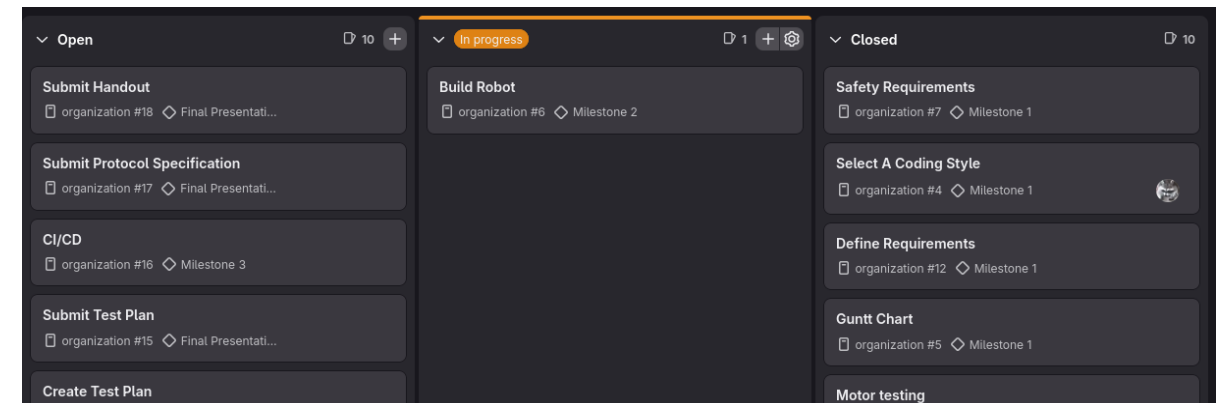
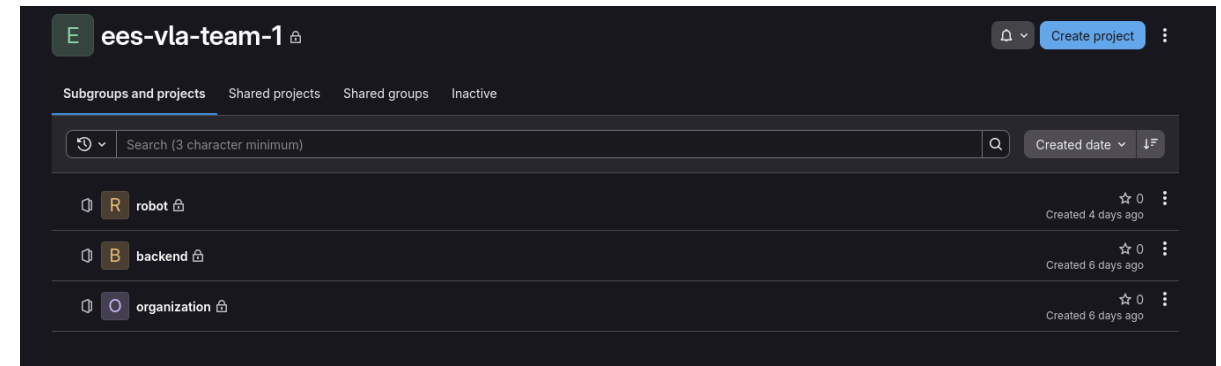
- Secure wiring to prevent mechanical stress or snagging.
- Mark robot's workspace. (range of the robot)
- Ensure fingers and hair stay clear of active joints.

- **Software (VLA):**

- Manual human override.
- Halt execution if prediction is harmful.
- Manually verify trajectory angles before execution.

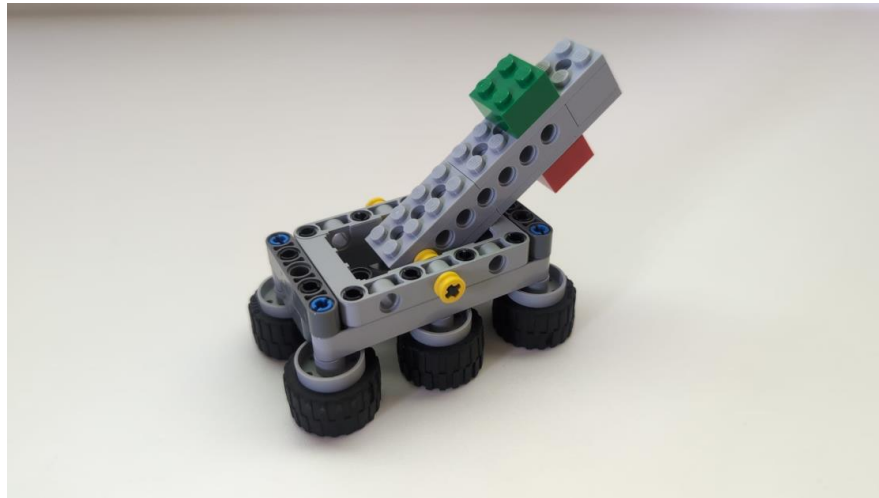
Results

- Two and a half meetings
- Created a git repository
- Organize the project
 - Project requirements
 - Project safety-requirements
 - Selected the programming language and style guides
- Started to test hardware and software
- Started to build with LEGO

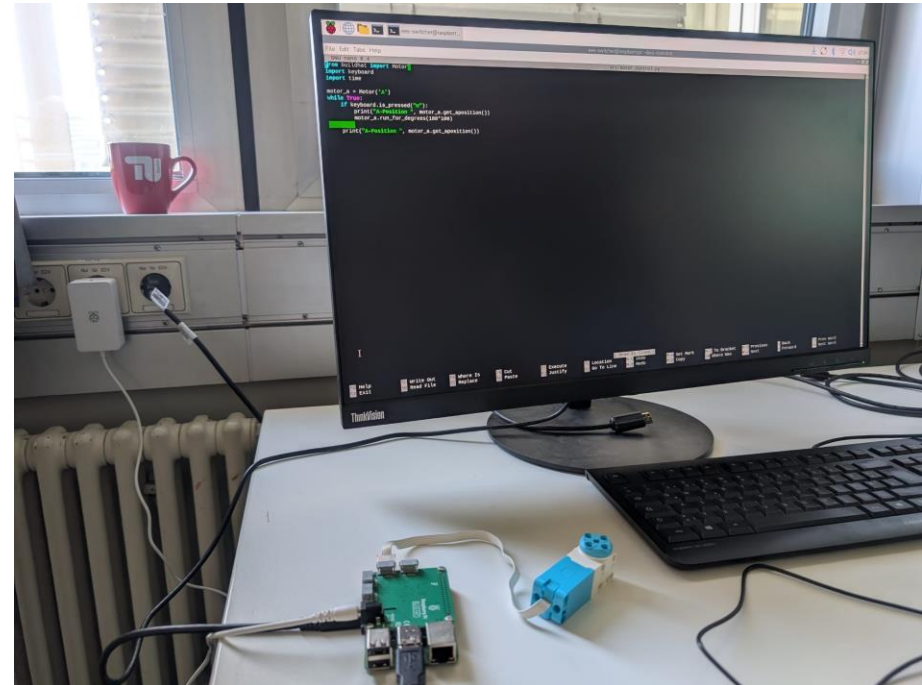


Git Repo: Projects and Issues

Results

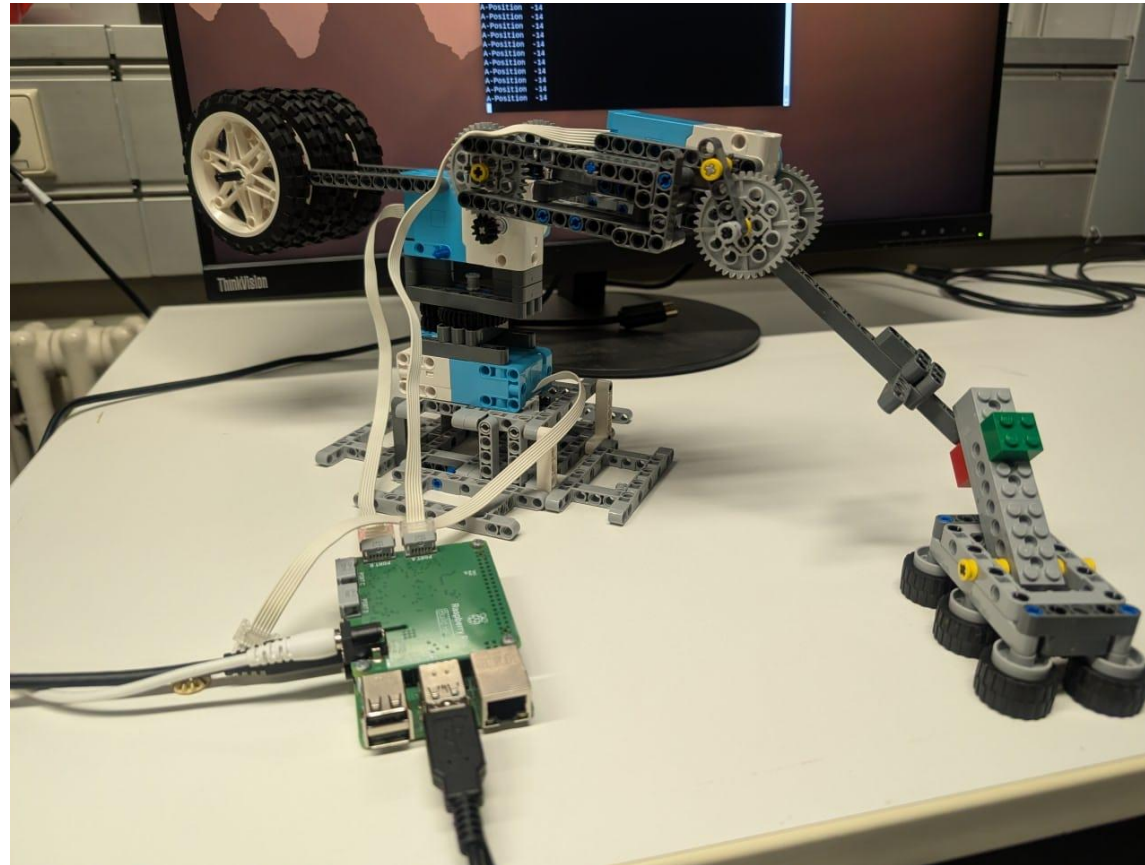


Build: LEGO lever



Software: tested motors

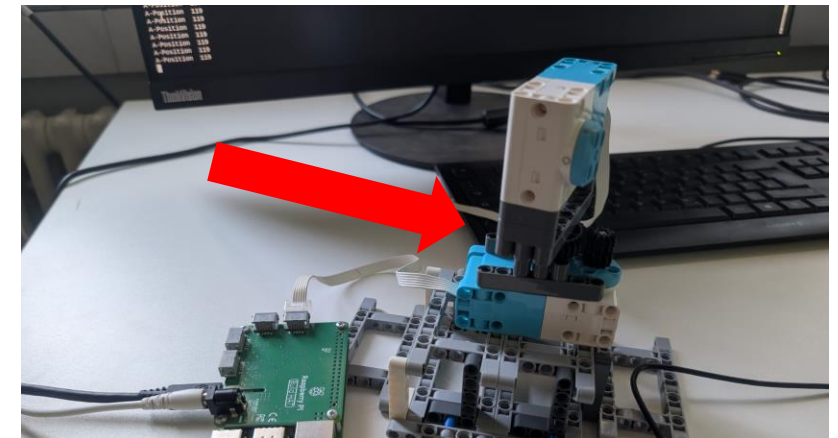
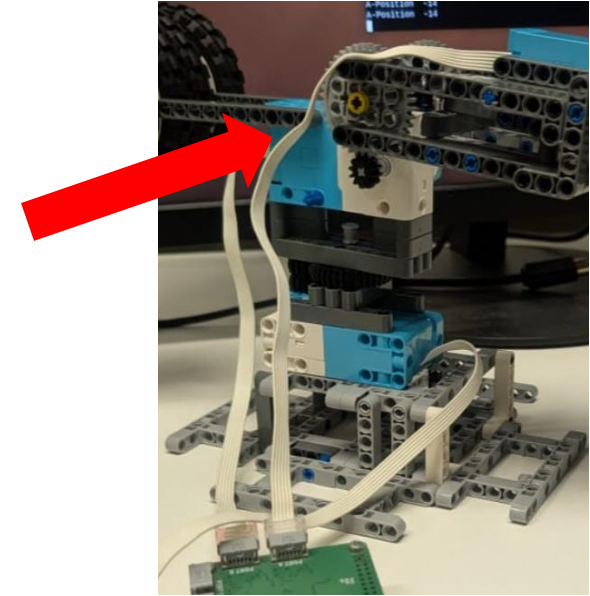
◆ Results - Prototype Robot



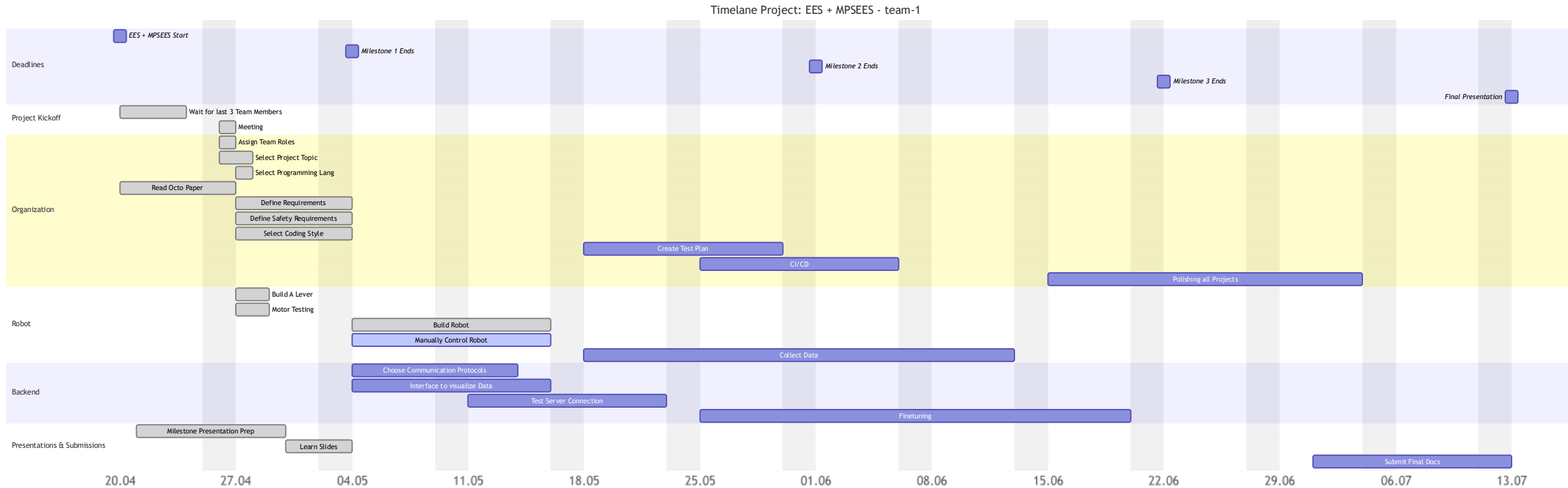
Build: LEGO robot arm

◆ Challenges

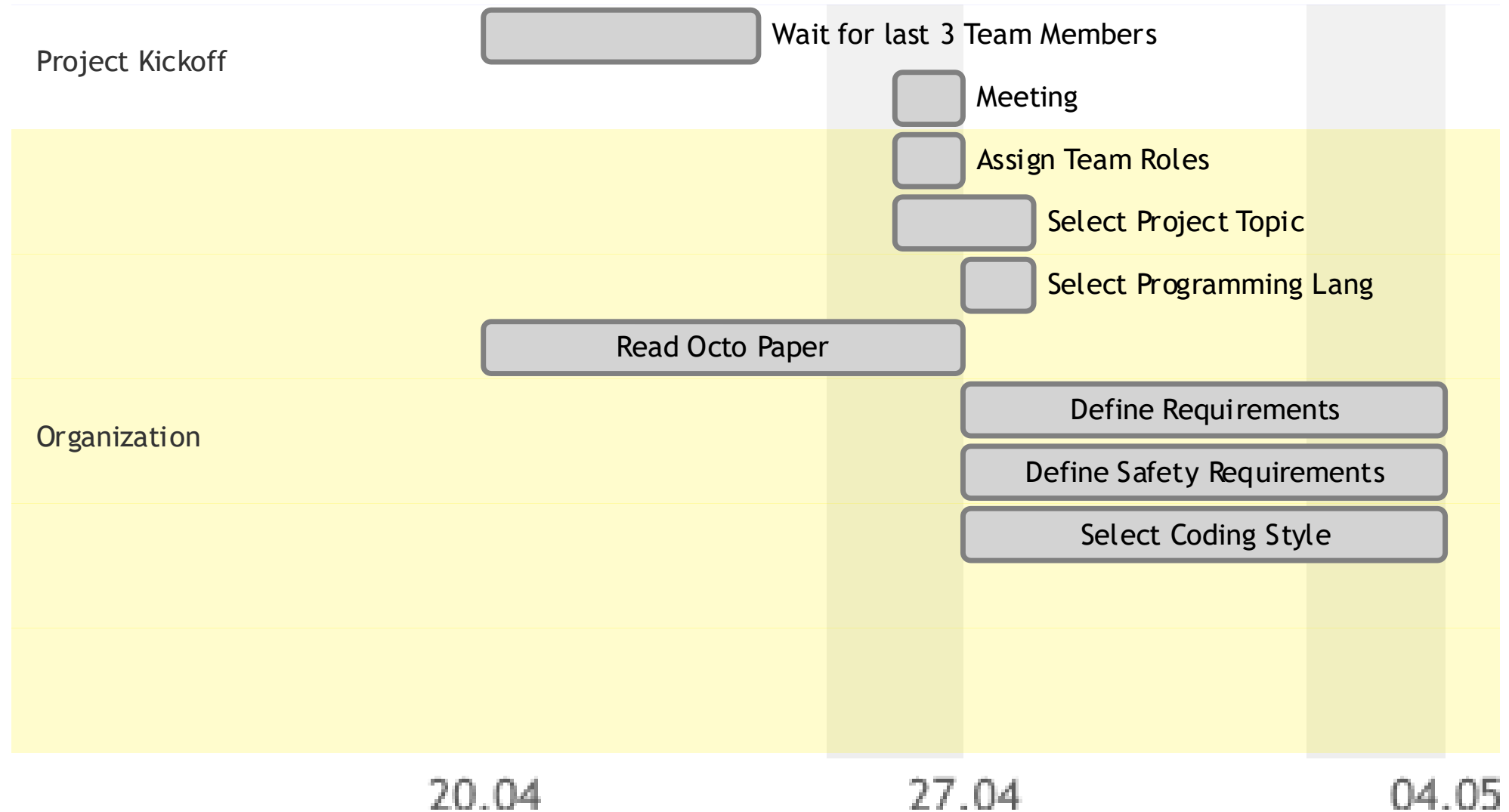
- Balance may need improvements
- Motor cable length at limit
- PID parameters may require further tuning
- Broken motor



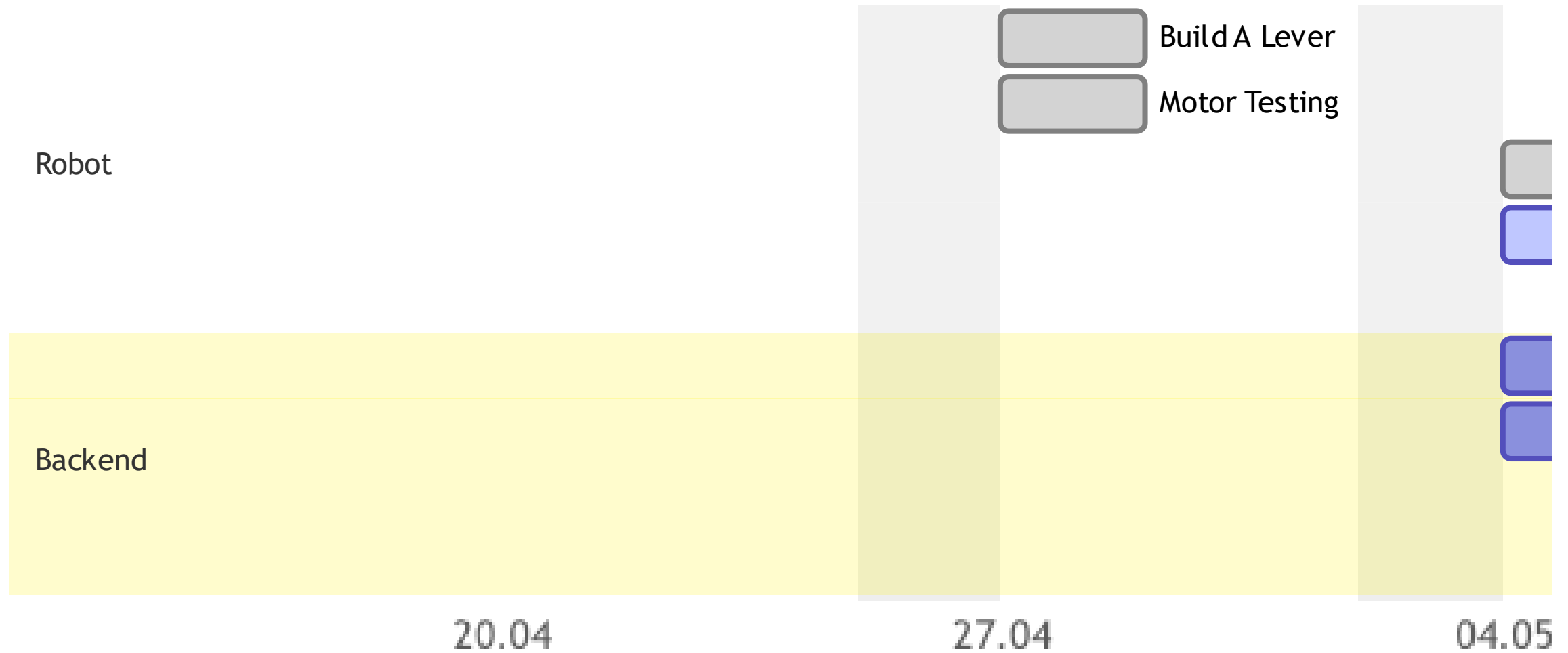
GANTT Chart



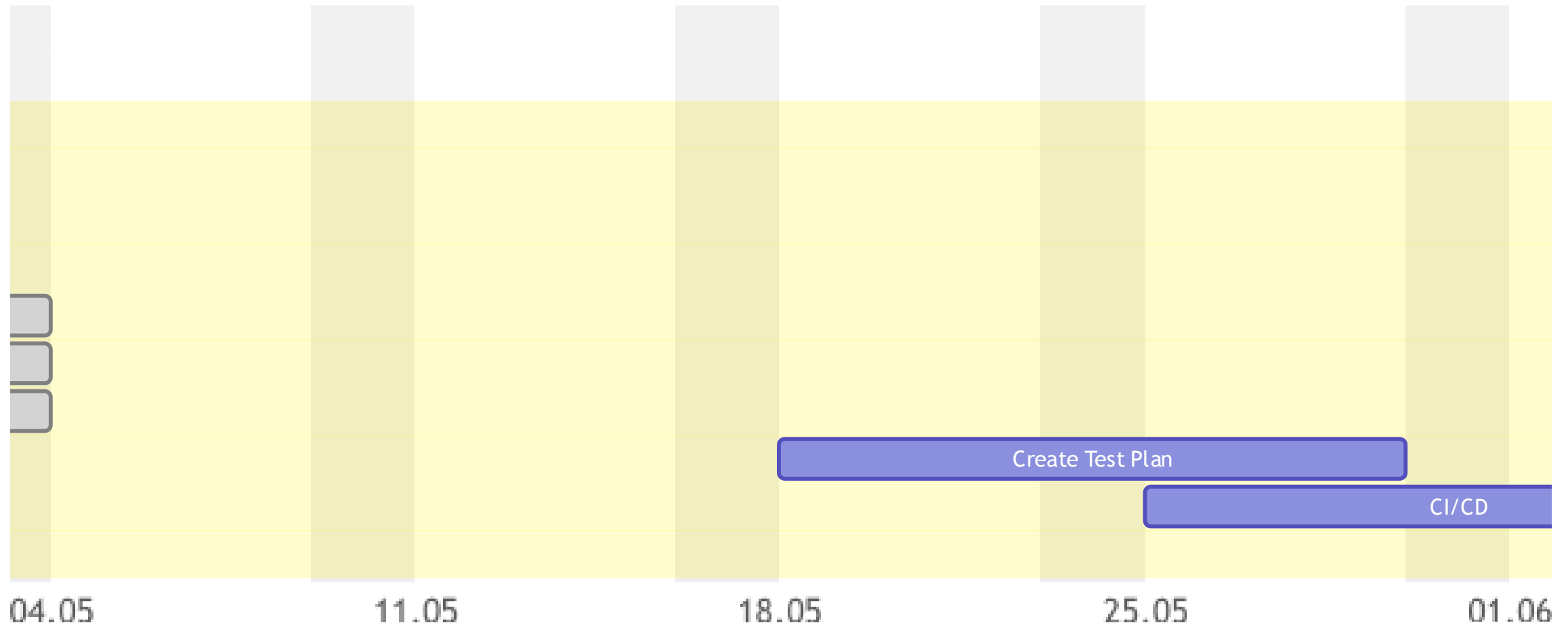
◆ GANTT Chart (Milestone 1)



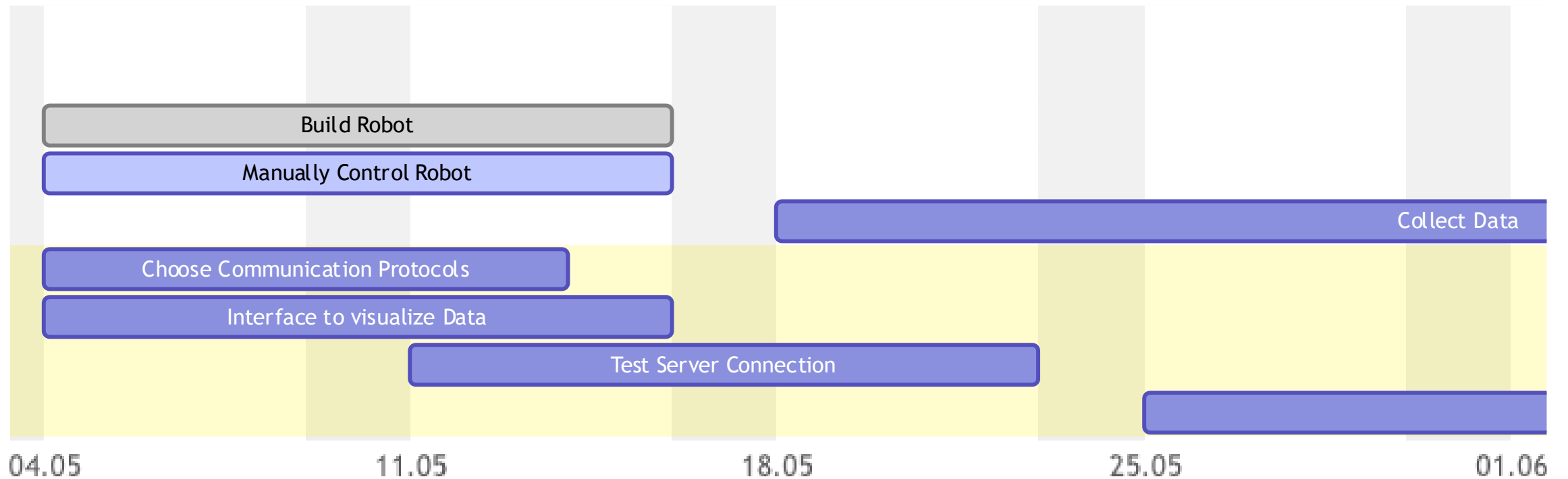
◆ GANTT Chart (Milestone 1)



◆ GANTT Chart (Planned for Milestone 2)



◆ GANTT Chart (Planned for Milestone 2)



◆ Reference List

- [1] Octo: <https://arxiv.org/pdf/2405.12213>
- [2] OpenVLA: <https://arxiv.org/pdf/2406.09246>
- [3] RT-2: <https://arxiv.org/pdf/2307.15818>
- [4] OtOaRoFM: <https://arxiv.org/pdf/2108.07258>
- [5] CLIP: <https://arxiv.org/pdf/2103.00020>
- [6] BridgeData V2: <https://arxiv.org/pdf/2308.12952>
- [7] Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware: <https://arxiv.org/pdf/2304.13705>
- [8] MathworksKinemaics: <https://www.mathworks.com/discovery/inverse-kinematics.html>
- [9] ISIS Kurs Entwurf Eingebetteter Systeme: <https://isis.tu-berlin.de/course/view.php?id=48172>

Questions?

